1. Accept a value x from user and use Linear search algorithm to check whether the number is present in the array or not and output the position if the number is present using python. (Using Function)

2. Accept a value x from user and use Binary search algorithm to check whether the number is present in the array or not and output the position if the number is present using python. (Using Function)

3. Write a Python program to sort a list of integers using the Bubble Sort algorithm.
   a. Accept n integer elements from the user.
   b. Sort the list in ascending order using the Bubble Sort technique.
   c. Display the original list and the sorted list.

4. Write a Python program to sort a list of integers using the Insertion Sort algorithm.
   a. Accept n integer elements from the user.
   b. Sort the list in ascending order using Insertion Sort.
   c. Display the original and sorted list.

5. Write a Python program to sort a list of integers using the Selection Sort algorithm.
   a. Accept n integer elements from the user.
   b. Sort the list in ascending order using Selection Sort.
   c. Display the original and sorted list.

6. Write a Python program to sort a list of numbers in ascending order using the Merge Sort algorithm.

7. Implement a dynamic singly linked list in Python to perform the following operations: Create, Insert, Display

8. Implement a dynamic singly linked list in Python to perform the following operations: Create, Delete, Display

9. Implement a dynamic singly linked list in Python to perform the following operations: Create, Display, and Reverse

10. Implement a dynamic singly linked list in Python to perform the following operations:
    a) Create a list

b) Search for a node by value
c) Display the list in forward direction

11. Write a Python program to implement a Doubly Linked List with the following operations:
    a) Insert a node at the end of the list
    b) Reverse the list
    c) Display the list in forward direction

12. Write a Python program to implement a Doubly Linked List with the following operations:
    a) Insert a node at the end of the list
    b) Delete a node at a given position
    c) Display the list in forward direction

13. Write a Python program to implement a Singly Circular Linked List with the following operations:
    a) Create a list (Insert a node at the end of the list)
    b) Display the list in forward direction

14. Write a Python program to implement a Doubly Circular Linked List with the following operations:
    a) Create a list (Insert a node at the end of the list)
    b) Display the list in forward direction

15. Create a Python program to implement a stack of integers using a fixed-size list (static implementation). The module should support the following operations:
    a) push(item) – Add an element to the top of the stack
    b) pop() – Remove and return the top element
    c) peek() – Return the top element without removing it
    d) display() – Show all elements in the stack

16. Create a Python program to implement a stack of integers using a linked list (dynamic implementation). The module should include the following operations:
    a) push(item) – Add an element to the top of the stack
    b) pop() – Remove and return the top element
    c) peek() – Return the top element without removing it
    d) display() – Show all elements in the stack

17. Write a Python program to implement Queue using Array (Static Implementation).

18. Write a Python program to implement Queue using Linked list (Dynamic Implementation)

19. Write a Python program to Implement a circular queue of integers implementation of the queue and implementing these five operations CreateQueue, Enqueue, Dequeue, Peek, IsEmpty (using static or a dynamic (circular linked list))

20. Write a Python program to implement reverse the string using stack.